

Updates on 3D OpenROAD Database

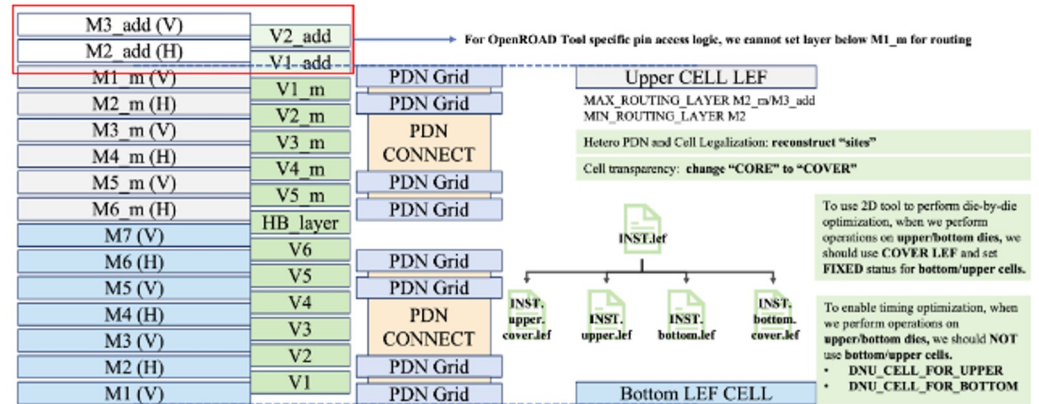
Osama Hammad
Cho Moon



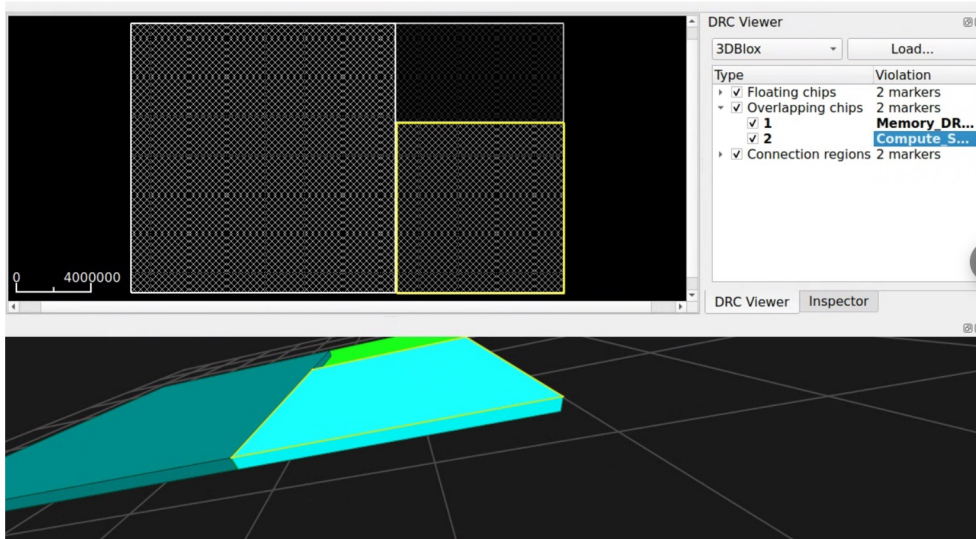
Precision Innovations is the primary industrial developer of OpenROAD and provides commercial support and customization for OpenROAD.

P&R for 3D IC Designs

- OpenROAD database (ODB) has been enhanced to read, write, visualize, analyze and optimize 3D designs
 - We want to invite more adoption of open-source tools and improve collaboration
- 3Dblox format is used for design I/O
- One design = multiple dies, multiple processes, bonded together
 - Each die/chiplet can have its own PDK and TechLEF
- ODB now models this end to end with persistency (write_db & read_db without data loss)
 - “Unfolded view” is available to enable full chip analysis for linting, visualization, timing analysis
- No more layer stack “hacks” for 3D designs

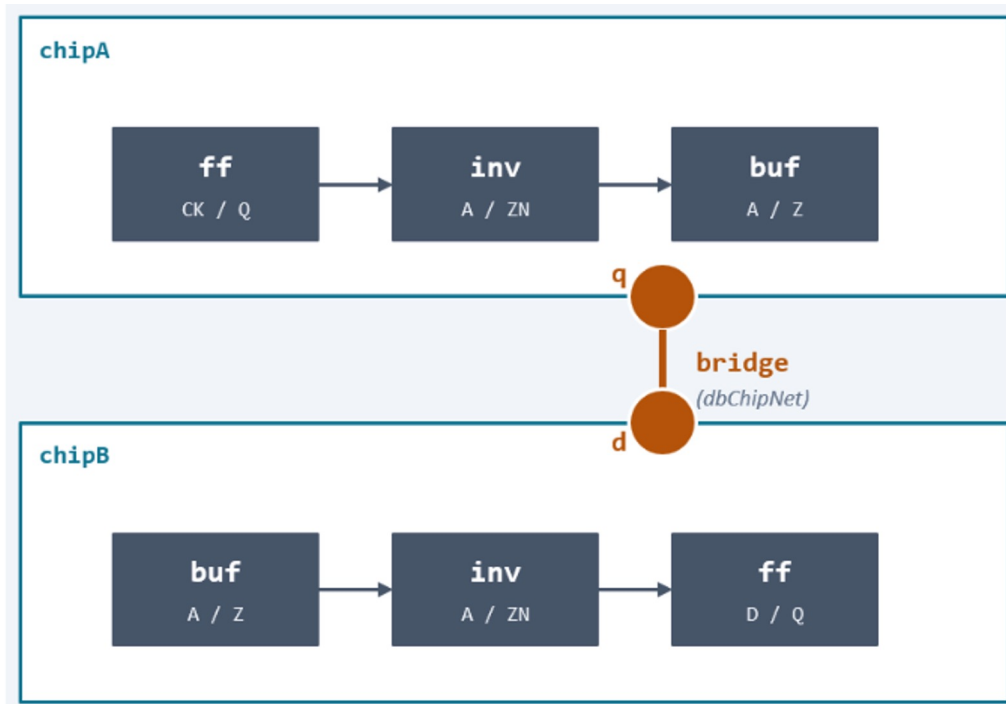


3D Viewer and Linter



Check	Purpose
Logical connectivity	Physically aligned bump pairs that belong to different nets.
Floating chips	Flags chip sets that aren't reachable to "ground". They're floating in the design
Overlapping chips	die chiplets whose 3D bounding boxes overlap.
Internal-ext usage	INTERNAL_EXT regions not referenced by any connection.
Connection regions	Connections whose regions don't xy-overlap or whose mating-surface gap mismatches the connection thickness.
Bump physical alignment	Bumps sitting outside their parent region's footprint.
Alignment markers	Missing, duplicate, or mis-oriented alignment markers between connected chips.

3D STA and RC Extractor



Delay	Time	Description

0.00	0.00	clock clk (rise edge)
0.00	0.00	clock network delay (ideal)
0.00	0.00	^ chipA/ff/CK (DFF_X1)
0.09	0.09	^ chipA/ff/Q (DFF_X1)
0.01	0.09	v chipA/inv/ZN (INV_X1)
0.02	0.11	v chipA/buf/Z (BUF_X1)
0.02	0.14	v chipB/buf/Z (BUF_X1)
0.01	0.15	^ chipB/inv/ZN (INV_X1)
0.00	0.15	^ chipB/ff/D (DFF_X1)
	0.15	data arrival time

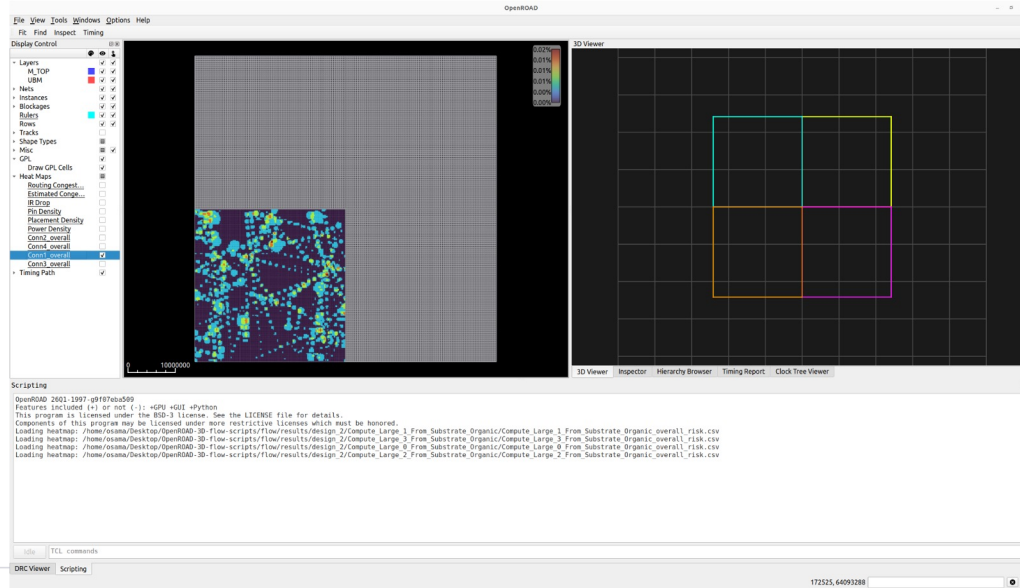
1.00	1.00	clock clk (rise edge)
0.00	1.00	clock network delay (ideal)
0.00	1.00	clock reconvergence pessimism
	1.00	^ chipB/ff/CK (DFF_X1)
-0.03	0.97	library setup time
	0.97	data required time

	0.97	data required time
-0.15		data arrival time

0.82		slack (MET)

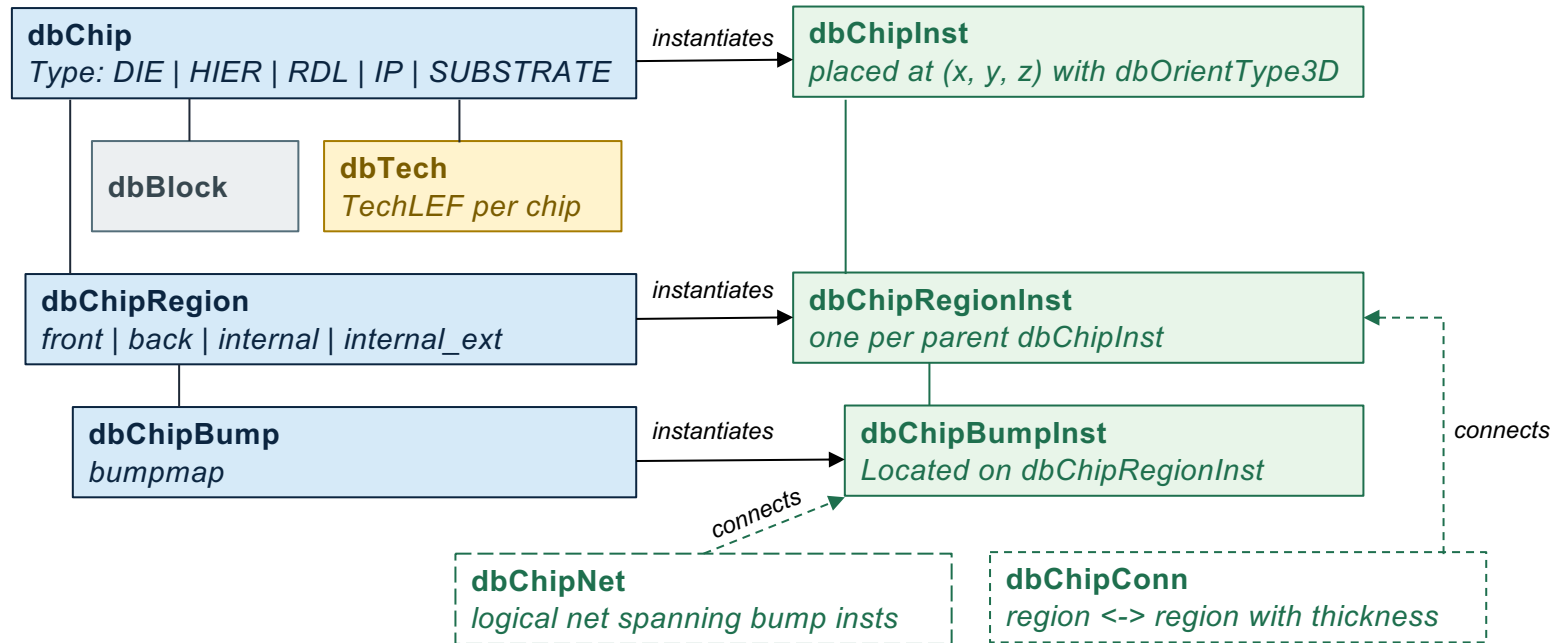
Yield Analysis

- ▶ YAP from UCLA has been integrated into OpenROAD via a file-based interface
 - ▶ Bond failure can be analysed
 - ▶ Heat map can be displayed



3D Database Overview

Same pattern as `dbMaster` / `dbInst`, lifted one level up to the chip itself.



3DBlox Files (3dbx and 3dbv)

read_3dbx top_design.3dbx

```
Design:
  name: top_design
  external:
    verilog_file: "top.v"
```

```
ChipletInst:
  cpu_inst:
    reference: cpu_complex
  ...
```

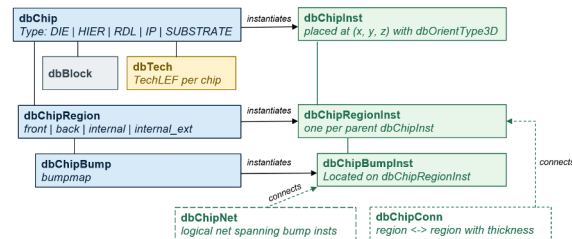
```
Stack:
  interposer_inst:
    loc: [0,0]
    z: 0.0
    orient: R0
  ...
```

```
Connection:
  Conn1:
    top: interposer_inst.regions.to_cpu
    bot: cpu_inst/compute_inst.regions.to_interposer
    thickness: 10.0
```

dbChip(HIER)

dbChipInst

dbChipConn



read_3dbv chiplet_definitions.3dbv

ChipletDef:

```
compute_tile:
  type: die
  design_area: [1000, 1000]
  thickness: 50
  regions:
    to_interposer:
      side: back
      coords: [[...]]
      bmap: compute_tile.bmap
  external:
    APR_tech_file: [tech_5nm.lef]
  cache_die:
    .....
```

dbChip

dbChipRegion

dbChipBump

dbTech

Additional Documentation

<https://github.com/The-OpenROAD-Project/OpenROAD/blob/master/src/odb/doc/3dic.md>